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HEALTH PACKAGE RECOMMENDATION SYSTEM

Prepared by- Aerisha Saxena (25BAI10017)



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COURSE- INTRODUCTION TO PROBLEM SOLVING AND PROGRAMMING

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INTRODUCTION

Healthcare affordability varies widely across income groups and age categories. People like senior citizens, middle-class families, and economically weaker sections—often struggle to identify suitable health check-up packages.

This project aims to simplify that choice by recommending an appropriate health package based on age and annual income, using a simple rule-based Python program.

PROBLEM STATEMENT

People belonging to different income and age groups have different healthcare needs.

For example:

- Poor families need essential tests at minimal cost.
- Middle-class individuals require affordable but comprehensive tests.
- Senior citizens need specialized screenings related to ageing.

There is no single system that suggests a suitable health package based on both age and income. This project solves that problem using a Health Package Recommendation System.

OBJECTIVES

The main objectives of this project are:

- To design a simple and effective health package recommendation system.
- To categorize users based on their age and income.
- To provide affordable health package suggestions.
- To apply Python program to solve real-life problems.
- To enhance understanding of structured programming and documentation.

FUNCTIONAL REQUIREMENTS

1. The system must take user inputs:

- Name
- Age
- Annual income

2. It must determine whether the user belongs to:

- Poor
- Middle-class
- Upper middle / Premium

3. It must recommend a health package based on:

- Age group
 - Income group
4. Display the selected health package clearly.
 5. Handle invalid inputs.

NON-FUNCTIONAL REQUIREMENTS

- Usability: Program should be easy and simple.
- Accuracy: Recommendations must follow predefined rules.
- Performance: The system must give instant output.
- Maintainability: Code should be readable and easy to understand.
- Portability: Should run on any OS supporting Python.

SYSTEM ARCHITECTURE

Input Layer → Processing Layer (health_package function) → Output Layer

- The user enters data.
- The system checks conditions using IF-ELSE logic.
- A suitable package is selected and displayed.

DESIGN DIAGRAM

- **Use case diagram**

Actor: User

Use Case Steps:

1. Enter details
2. System processes input
3. System selects package
4. System displays recommended package

- **Workflow diagram:**

1. Start
2. Input- name, age, income
3. Check income and age group
4. Determine package
5. Display recommended package
6. End

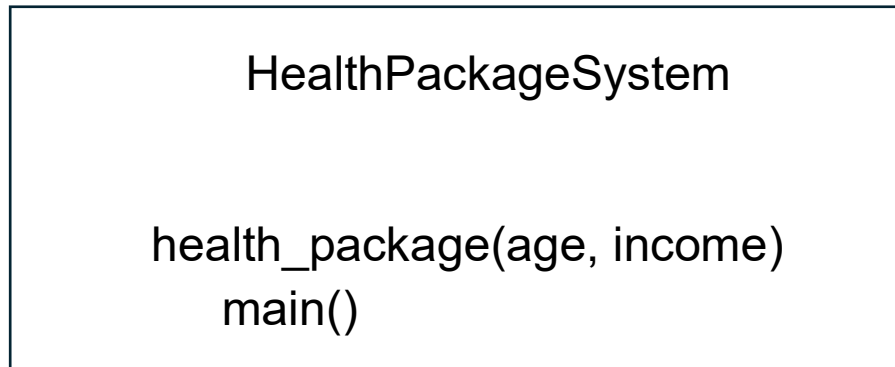
- **Sequence Diagram**

User → System

1. Enter name
2. Enter age
3. Enter income
4. Input sent to function

5. health_package() evaluates
6. System returns package
7. Display to user

- **Class/Component Diagram**



DESIGN DECISIONS & RATIONALE

- If–else decision structure chosen for simplicity.
- (Age + Income) determine health packages most effectively.
- Outputs are displayed in multi-line text for readability.
- No database used to keep the project simple.
- Function-based structure for modularity and easy updates.

IMPLEMENTATION DETAILS

- Language: Python
- Concepts used:

- Input handling
 - Data type conversion
 - Conditional branching
 - Modular function design
- Main logic implemented in `health_package()`

SCREENSHOTS

```
# Health Package Program for Middle-Class, Old-Age, and Poor People

def health_package(age, income):
    # Poor People Package
    if income < 50000:
        if age >= 60:
            return """
--- Poor Senior Citizen Health Package ---
• Blood Pressure Check
• Sugar Test
• Anaemia Test
• Free BP Test
Cost: ₹0 - ₹150
"""
        else:
            return """
--- Poor Family Health Package ---
• CBC
• Blood Sugar
• Quick Kidney/Liver Test
• Urine Routine
Cost: ₹200 - ₹500
"""

    # Middle-Class Packages
    if 50000 <= income <= 300000:
        if age >= 60:
            return """
```

```

        --- Middle-Class Senior Health Package ---
        • ECG
        • Diabetes Tests (FBS, PP, HbA1c)
        • Kidney & Liver Function
        • Bone Density Screening
        Cost: ₹1500 – ₹2500
        """
    else:
        return ""

    --- Middle-Class Family Package ---
    • CBC
    • Thyroid Test (T3, T4, TSH)
    • Kidney & Liver Function
    • Eye & Dental Check-up
    • Vitamin D & B12
    Cost: ₹3000 – ₹5000
    """

    # Richer Middle/Upper Class Packages
    if income > 300000:
        if age >= 60:
            return ""

        --- Premium Senior Citizen Package ---
        • 2D Echo + ECG
        • Complete Body Check-up
        • Dementia Screening
        • Bone Health & Vitamin Tests
        Cost: ₹5000 – ₹8000
        """

    else:
        return ""

    --- Premium Lifestyle Package ---
    • Cardiac Markers
    • Thyroid Panel
    • Liver, Kidney, Lipid Profile
    • Diet & Fitness Consultation
    Cost: ₹4000 – ₹6500
    """

    return "Invalid Input"

print("Health Package Recommendation System")
name = input("Enter your name: ")
age = int(input("Enter your age: "))
income = int(input("Enter your annual income (in ₹): "))

package = health_package(age, income)

print("Hello", name, "!")
print("Based on your age and income, your recommended health package is:")
print(package)

```

```

Health Package Recommendation System
Enter your name: Adrija
Enter your age: 70
Enter your annual income (in ₹): 60000
Hello Adrija !
Based on your age and income, your recommended health package is:

```

```

--- Middle-Class Senior Health Package ---
• ECG
• Diabetes Tests (FBS, PP, HbA1c)
• Kidney & Liver Function
• Bone Density Screening
Cost: ₹1500 – ₹2500

```

TESTING APPROACH

Black box testing

Test No	Age	Income	Expected Output
1	24	20000	Poor Family Package
2	68	30000	Poor Senior Package
3	39	120000	Middle-Class Family
4	71	180000	Middle-Class Senior
5	20	600000	Premium Lifestyle
6	80	700000	Premium Senior

CHALLENGE FACED

- Ensuring no overlap between income ranges.
- Maintaining proper indentation in Python.
- Formatting readable multi-line output.
- Designing logical decision flow for all combinations.

LEARNING AND KEY TAKEAWAYS

- Learned how to build rule-based systems.
- Improved Python conditional logic skills.
- Understood how to write structured documentation.
- Gained confidence in modular programming.

- Applied real-life reasoning to technical implementation.

FUTURE ENHANCEMENTS

- Add gender, medical history, lifestyle factors.
- Provide hospital recommendations based on location.

REFERENCES

<https://www.geeksforgeeks.org/system-design/unified-modeling-language-uml-introduction/>

https://colab.research.google.com/drive/1D2erxT6_Bq6yhwx8watO0bKTd4C-NGdP